

Mousumi Maitra

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Current Position

Post Doctoral Research Fellow
January, 2025– Present
Dr. Arpan Bhattacharyya
Department of Physics
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Personal Details

D.O.B. **21.12.1991.**
Marital .
Status **Married.**
Religion **Hindu.**
Permanent Address **B18, Nalanda apartment, IIT Delhi campus, Hauz Khas, New Delhi-110016.**

Educational Details

July,2016– June,2022 **Ph.D, PhD viva-voce date-21.06.2022**, Indian Institute of Technology Guwahati, Guwahati, India.
May,2016 **M.Sc in physics**, *University of Calcutta*, Kolkata, India, *Percentage of Marks-69.5.*
1st class
July,2014 **B.Sc in physics**, *University of Calcutta*, *Percentage of marks-73.5.*
1st class

CSIR-NET, GATE qualification

CSIR-NET **2016 JUNE qualified**, *all India Rank, LS 60 .*
GATE-2016 **qualified.**

Ph.D Thesis

- **Title:** *Asymptotic symmetry and its role in black hole thermodynamics*
- **Supervisors :** Dr. Debaprasad Maity and Dr. Bibhas Ranjan Majhi

Current Research Interest and ongoing research work

- **Next to MHV level scattering amplitudes in Celestial holography**

- Worldline QFT techniques for the computation of gravitational waves observables for theories beyond GR
- Gravitational self force computation for plunge orbit

--- Ongoing Interdisciplinary project in collaboration with IIT Delhi

- Quantum Walk Markov Process and its application to various physical systems.

--- Post-Doctoral Experience

Post Doctoral Fellow Dec,2022–Dec,2024, *Mentor : Dr. Shamik Banerjee*, School of Physical Sciences, National Institute of Science Education and Research, Bhubaneswar.

--- Teaching experience during PhD time

B.Tech Lab class (winter 2017 sem- monsoon 2018 sem, IIT Guwahati)

M.sc 2ns sem lab class (winter 2019 Semester, IIT Guwahati)

M.sc 4th sem (monsoon 2019 Sem-winter 2020 sem, IIT Guwahati)

--- Thesis supervision during Postdoc time

June, 2025-May, 2026 **co-supervised one M.Tech student**, from *Department of Mathematics*, IIT Delhi, Topic: Quantum Walk Markov Process and its application to various physical systems.

--- Published or Communicated Papers

- Shamik Banerjee, Mousumi Maitra, Raju Mandal, Milan Patra, **Holographic symmetry algebra for the MHV sector revisited**, (*JHEP* 12 (2025) 175)
- Mousumi Maitra, Debaprasad Maity and Bibhas Ranjan Majhi, **BMS Goldstone modes near the horizon of a Kerr black hole are thermal**, 2022, (*Phys.Lett.B* 824 (2022) 136825)
- Mousumi Maitra, Debaprasad Maity and Bibhas Ranjan Majhi, **Symmetries near a generic timelike surface in black hole spacetime**, (*Class. Quant. Grav.* 38 (2021) 14, 145027)
- Mousumi Maitra, Debaprasad Maity and Bibhas Ranjan Majhi, **Near horizon symmetries, emergence of Goldstone modes and thermality**, (*Eur. Phys. J. Plus* 135 483 (2020))
- Mousumi Maitra, Debaprasad Maity and Bibhas Ranjan Majhi, **Symmetries near a generic charged null surface and associated algebra: An off-shell analysis** (*Phys.Rev.D* 97 (2018) 12, 124065)

--- iNSPIRE Link

<https://inspirehep.net/authors/1656815>.

--- Awards/Recognition

2011-2016 **DST-INSPIRE scholarship by govt. of India**, Top 1 percentage rank holder in the 10+2 examination.

August, 2019 **Mukul Bose Memorial Prize, 2015 by the University of Calcutta**, MSc 1st class.

Conferences / seminars attended recently

- Oct, 2025 **International meeting titled '10 Years of Gravitational Wave Discovery: Advancing the Frontiers of Gravitational Physics', IIT Gandhinagar.**
- April, 2023 **String theory seminar, NISER, Bhubhaneswar.**
- Dec, 2023 **Annual String meetings, IIT Bombay.**
- Oct, 2021 **AAPPS-DACG Workshop 2021 on Astrophysics, Cosmology and Gravitation, Online seminar, Asia Pacific Center for Theoretical Physics, Korea.**
- Dec, 2020 **"Quasinormal modes from black holes", 31th meeting of Indian Association for General Relativity and Gravitation (IAGRG), Online seminar, IIT Gandhinagar, India**
- February, 2019 **2019 YITP Asian-Pacific Winter School and Workshop on Gravitation and Cosmology, at YITP, Kyoto, Japan.**
- January, 2019 **"New Eras in Cosmology and Multi-messenger Astronomy", 30th meeting of Indian Association for General Relativity and Gravitation (IAGRG), at BITS, Pilani Hyderabad campus, India.**
- February, 2018 **Research Conclave, IIT Guwahati, Assam, India.**
- Dec, 2017 **Black Holes: From classical to quantum gravity, at IIT Gandhinagar, Gujarat, India.**
- May, 2017 **The Era of Gravitational Waves, IAGRG-29, at IIT Guwahati.**

Conference Presentation

- Oct, 2021 **Oral presentation (online) on "Near horizon symmetries, emergence of Goldstone modes and thermality", at AAPPS-DACG Workshop 2021 on Astrophysics, Cosmology and Gravitation, Asia Pacific Center for Theoretical Physics, Korea.**
- February, 2019 **Postar presentation on "Symmetries near a generic charged null surface and associated algebra: An off-shell analysis", at 2019 YITP Asian-Pacific Winter School and Workshop on Gravitation and Cosmology, YITP, Kyoto, Japan.**
- January, 2019 **Oral presentation on "Symmetries near a generic charged null surface and associated algebra: An off-shell analysis", at 30th meeting of Indian Association for General Relativity and Gravitation (IAGRG), at BITS, Pilani Hyderabad campus, India.**

Languages

English	Read, Write and Speak
Hindi	Read, Write and Speak
Bengali	Read, Write and Speak

Mother Tongue

Interests

Verbal and non verbal reasoning solve

Collect Post stamp

References

Prof. Shamik Banerjee

School of Physical Sciences
NISER, Bhubhaneswar, India
Jatni, India, 782050

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**Dr. Bibhas Ranjan Majhi**

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